

Ê PHASOR TEMPLATES (Air:  $\epsilon_r = \mu_r = 1$ )

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega \sqrt{\epsilon_{r,i}}}{c} \text{ (rad/m)} \Rightarrow k_2 = k_1 \sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$$

Direction table  $u \in \{x, y, z\} = \text{axis perp. to boundary}$

| Inc dir    | Inc           | Refl          | Trans         |
|------------|---------------|---------------|---------------|
| $+\hat{u}$ | $e^{-jk_1 u}$ | $e^{+jk_1 u}$ | $e^{-jk_2 u}$ |
| $-\hat{u}$ | $e^{+jk_1 u}$ | $e^{-jk_1 u}$ | $e^{+jk_2 u}$ |

$+\hat{u}$  direction  $\Leftrightarrow$  Med 2 at  $u \geq 0$ . Reflected sign flips, transmitted matches incident.

General template (incident has pol.  $\hat{e}$ , amplitude  $a_0$ ):

$$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u} \quad (\text{incident wave, Med 1})$$

$$\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u} \quad (\text{reflected wave, Med 1})$$

$$\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u} \quad (\text{transmitted wave, Med 2})$$

(lossy med 2: replace  $e^{\mp j k_2 u}$  with  $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$ )

Total in medium 1:  $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$ .

$\hat{e}$  by polarization *plug into templates above*

| Pol.               | $\hat{e}$                               | Note              |
|--------------------|-----------------------------------------|-------------------|
| Lin. $\hat{x}$     | $\hat{x}$                               | $E_y = 0$         |
| Lin. $\hat{y}$     | $\hat{y}$                               | $E_x = 0$         |
| Lin. at $45^\circ$ | $(\hat{x} + \hat{y})/\sqrt{2}$          | in phase          |
| RHC                | $\hat{x} - j\hat{y}$                    | $\delta = -\pi/2$ |
| LHC                | $\hat{x} + j\hat{y}$                    | $\delta = +\pi/2$ |
| General            | $a_x \hat{x} + a_y e^{j\delta} \hat{y}$ | elliptical        |

$\Gamma, \tau$  act on  $\hat{e}$  unchanged. Only swap the polarization, not the formulas.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted  *$\eta$ -ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check:  $\%P_r + \%P_t = 100$ .

If medium 2 is denser ( $\eta_2 < \eta_1$ ),  $|\tau| > 1$  but  $\%P_t < 100\%$  because of the  $\eta_1/\eta_2$  factor.

LOSSY MEDIA — 5 CASES

Loss tangent  $\epsilon' = \epsilon_r \epsilon_0$ ,  $\epsilon_0 \approx 1/(36\pi \times 10^9)$

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$$

| Type         | $\sigma/\omega\epsilon$ | $\eta_c$ , $\alpha$ , $\beta$                                                                                                                                                                                       |
|--------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lossless     | 0                       | $\eta = \eta_0/\sqrt{\epsilon_r}$ exact; $\alpha = 0$ , $\beta = \omega\sqrt{\epsilon_r}/c$                                                                                                                         |
| Low-loss     | $< 10^{-2}$             | $\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right)$ ; $\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}$ , $\beta \approx \frac{\omega\sqrt{\epsilon_r}}{c}$ |
| Quasi-cond.  | $10^{-2}$ – $10^2$      | exact: $(\eta/\sqrt{X})e^{j\theta\eta}$ , see below                                                                                                                                                                 |
| Good cond.   | $> 10^2$                | $\eta_c \approx (1+j)\sqrt{\pi f \mu/\sigma}$ ; $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$                                                                                                              |
| Magnetic any |                         | keep full $\sqrt{\mu_r/\epsilon_r} \eta_0$                                                                                                                                                                          |

For  $\Gamma/\tau$ : drop  $j$  correction; use  $\eta_0/\sqrt{\epsilon_r}$ . Magnetic: keep  $\mu_r$ .

Exact (quasi-cond)  $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$

$$\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X - 1}$$

$$\beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X + 1}$$

$$\theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$$

BOUNDARY

$\eta_1, \eta_2$  — imp. ( $\Omega$ )  
 $\Gamma$  — refl. coeff (unitless)  
 $t_i = 1 + \Gamma$  — trans. coeff  
 $k_i = \omega \sqrt{\epsilon_{r,i}}/c$  (rad/m)  
 $\alpha_2, \beta_2$  — if med 2 lossy  
 $z = 0$  — boundary plane  
 $\eta_0 = 120\pi \approx 377 \Omega$

SNELL / OBLIQUE

$\theta_i, \theta_t$  — angles from normal  
 $n_i = \sqrt{\epsilon_{r,i}}$  — index of refr.  
 $t_i$  — slab thickness (m)  
 $d$  — lateral disp. (m)  
plane of inc. = plane of  $\hat{k}$ +normal  
 $\perp$ :  $\mathbf{E} \perp$  this plane  
 $\parallel$ :  $\mathbf{E} \parallel$  this plane

WAVEGUIDE

$a, b$  — wide, narrow dim. (m);  
 $a > b$   
 $m, n$  — mode indices (int.)  
 $f_{mn}$  — cutoff (Hz)  
 $\mu$  — prop. const (rad/m)  
 $Z_{TE/TM}$  — wave imp. ( $\Omega$ )  
 $E_x, H_y$  — field comps (V/m, A/m)  
 $\eta = \eta_0/\sqrt{\epsilon_r}$

VELOCITIES

$u_{p0} = c/\sqrt{\epsilon_r}$  — bulk phase vel.  
 $u_p = u_{p0}/\sqrt{1 - (f_c/f)^2}$   
 $u_g = u_{p0}\sqrt{1 - (f_c/f)^2}$   
 $u_p u_g = u_{p0}^2$   
 $t = L/u_g$  — travel time  
 $c = 3 \times 10^8$  m/s

POWER & UNITS

$\%P_r = 100|\Gamma|^2$   
 $\%P_t = 100|\tau|^2 \eta_1/\eta_2$   
 $\%P_r + \%P_t = 100$   
 $\sigma$  (S/m),  $\epsilon$  (F/m),  $\mu$  (H/m)  
 $\epsilon_0 = 8.85 \times 10^{-12}$  F/m  
 $\mu_0 = 4\pi \times 10^{-7}$  H/m  
 $1/(36\pi) \times 10^{-9} \approx \epsilon_0$

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r,1}}{\epsilon_{r,2}}} = \sin \theta_1 \frac{n_1}{n_2}$$

Multilayer chain *stack of slabs*  $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

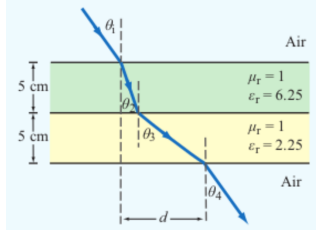
$$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

*Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).*

LATERAL DISPLACEMENT



Beam offset through  $N$  slabs thickness  $t_i$ , refracted angle inside slab  $i$

Slab-indexed  $\theta_i$  = angle inside slab  $i$

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

Angle-indexed (HW3 P3 style)  $\theta_1$  = incident,  $\theta_{i+1}$  = inside slab  $i$

$$d = \sum_{i=1}^N t_i \tan \theta_{i+1}$$

Same physics, different numbering. In HW3 P3: slab 1 contributes  $t_1 \tan \theta_2$ , slab 2 contributes  $t_2 \tan \theta_3$ . The angle in the formula is always the one the ray makes inside that slab.

Air–slab–air: exit angle returns to  $\theta_1$  but is laterally shifted by  $d$ .

TABLE 8-2 —  $\Gamma, \tau, R, T$  SUMMARY

Med. 1 ( $z < 0$ ) incident side; Med. 2 ( $z \geq 0$ ) transmitted. Shorthand:  $c_i = \cos \theta_i$ ,  $c_t = \cos \theta_t$ .

|                     | Normal ( $\theta_i = 0$ )                 | $\perp$ pol                                               | $\parallel$ pol                                           |
|---------------------|-------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|
| $\Gamma$ refl. wave | $\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$ | $\frac{\eta_2 c_i - \eta_1 c_t}{\eta_2 c_i + \eta_1 c_t}$ | $\frac{\eta_2 c_t - \eta_1 c_i}{\eta_2 c_t + \eta_1 c_i}$ |
| $\tau$ trans. wave  | $\frac{2\eta_2}{\eta_2 + \eta_1}$         | $\frac{2\eta_2 c_i}{\eta_2 c_i + \eta_1 c_t}$             | $\frac{2\eta_2 c_t}{\eta_2 c_t + \eta_1 c_i}$             |
| $\tau$ rel.         | $\tau = 1 + \Gamma$                       | $\tau_\perp = 1 + \Gamma_\perp$                           | $\tau_\parallel = (1 + \Gamma_\parallel) \frac{c_i}{c_t}$ |
| $R$ refl. pwr       | $ \Gamma ^2$                              | $ \Gamma_\perp ^2$                                        | $ \Gamma_\parallel ^2$                                    |
| $T$ trans. pwr      | $ \tau ^2 \frac{\eta_1}{\eta_2}$          | $ \tau_\perp ^2 \frac{\eta_1 c_t}{\eta_2 c_i}$            | $ \tau_\parallel ^2 \frac{\eta_1 c_t}{\eta_2 c_i}$        |
| $R+T$ total         | $T = 1 - R$                               | $T_\perp = 1 - R_\perp$                                   | $T_\parallel = 1 - R_\parallel$                           |

Notes:  $\sin \theta_t \approx \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$ ; nonmag.  $\Rightarrow \eta_2/\eta_1 = n_1/n_2$ .

Intrinsic impedance  $\eta_i$  *plug into table above*

$$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\epsilon_{r,i}} \text{ ( $\Omega$ )} \quad \eta_0 = 120\pi \approx 377 \Omega$$

Default  $\mu_r = 1$  (air, dielectric). Use  $\mu_r \neq 1$  only if problem says “ferromagnetic”/iron/ferrite/etc.

Low-loss correction  $\eta_c \sigma/(\omega\epsilon) \ll 1$

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$$

For  $\Gamma/\tau$  just use  $\eta_0/\sqrt{\epsilon_r}$ . The  $j$  term is for  $|\eta_c|, \theta_\eta$ . See LOSSY MEDIA — 5 CASES for all regimes.

RECTANGULAR WAVEGUIDE — MODES

TE mode  $E_z = 0, H_z \neq 0$

TM mode  $H_z = 0, E_z \neq 0$

Dimensions  $a \times b$  with  $a > b$  along  $\hat{x}, \hat{y}$ ; wave in  $\hat{z}$ .

$E_x$  form (TE) *read  $m, n$  from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on  $x = m\pi/a$ ; coef. on  $y = n\pi/b$ .

Identify  $\text{TE}_{mn}$  vs  $\text{TM}_{mn}$  *cos/sin pattern of  $E_x$  is identical for both!*

1. Read the problem statement (e.g. “a TE wave”  $\rightarrow$  TE).  
2. If  $m=0$  or  $n=0$  in the mode label  $\Rightarrow$  must be TE (TM forbids 0).

3. Source field  $H_z(x, y)$  given  $\Rightarrow$  TE. Source field  $E_z(x, y)$  given  $\Rightarrow$  TM.

TE allows at least one of  $m, n \geq 1$  (other can be 0). TM needs both  $m, n \geq 1$ . That’s why  $\text{TE}_{10}$  exists but  $\text{TM}_{10}$  doesn’t.

TABLE 8-3 — FULL FIELD COMPONENTS

Shorthand:  $C_X = \cos \frac{m\pi x}{a}$ ,  $S_X = \sin \frac{m\pi x}{a}$ ,  $C_Y = \cos \frac{n\pi y}{b}$ ,  $S_Y = \sin \frac{n\pi y}{b}$ . All fields carry an  $e^{-j\beta z}$  factor (omitted below).  $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$ .

TE mode *generator:  $\hat{H}_z$*

$$\hat{H}_z = H_0 C_X C_Y$$

$$\hat{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 C_X S_Y$$

$$\hat{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 S_X C_Y$$

$$\hat{E}_z = 0, \quad \hat{H}_x = -\hat{E}_y/Z_{\text{TE}}, \quad \hat{H}_y = \hat{E}_x/Z_{\text{TE}}$$

TM mode *generator:  $\hat{E}_z$*

$$\hat{E}_z = E_0 S_X S_Y$$

$$\hat{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 C_X S_Y$$

$$\hat{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 S_X C_Y$$

$$\hat{H}_z = 0, \quad \hat{H}_x = -\hat{E}_y/Z_{\text{TM}}, \quad \hat{H}_y = \hat{E}_x/Z_{\text{TM}}$$

TEM (plane wave) for reference:  $\hat{E}_x = E_0 e^{-j\beta z}$ ,  $\hat{H}_y = \hat{E}_x/\eta$ ,  $f_c$  N/A,  $k = \omega\sqrt{\mu\epsilon}$ .

CUTOFF FREQUENCY

$f_{mn}$   $m, n$  integers  $\geq 0$  (not both zero)

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

Common cutoffs ( $a > b$ )

| Quantity                         | Formula                                       |
|----------------------------------|-----------------------------------------------|
| $u_{p0}$                         | $c/\sqrt{\epsilon_r}$ (hollow: $u_{p0} = c$ ) |
| $\text{TE}_{10}$                 | $u_{p0}/(2a)$ (dominant)                      |
| $\text{TE}_{01}$                 | $u_{p0}/(2b)$                                 |
| $\text{TE}_{20}$                 | $u_{p0}/a = 2f_{10}$                          |
| $\text{TE}_{11}, \text{TM}_{11}$ | $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$       |

Propagation requires  $f > f_{mn}$ .

$\epsilon_r$  FROM DISPERSION

Given  $\omega, \beta$ , mode ( $m, n$ )

$$\epsilon_r = \frac{c^2}{\omega^2} \left[ \beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$$

Sanity:  $\beta$  in rad/m,  $a, b$  in m,  $\omega$  in rad/s.

WAVE IMPEDANCE &  $H_y$

$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \text{ ( $\Omega$ )} \quad Z_{\text{TM}} = \eta \sqrt{1 - (f_c/f)^2}$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{\text{TE}}} \text{ (TE mode)}$$

$Z_{\text{TE}} > \eta$  always (denominator  $< 1$ );  $Z_{\text{TM}} < \eta$ .

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2}$  (m/s)  
 $u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}}$  (m/s)  $\Rightarrow u_p u_g = u_{p0}^2$  (m<sup>2</sup>/s<sup>2</sup>)  
 $t = L/u_g$  (s) travel time through guide of length  $L$   
Higher modes have lower  $u_g \Rightarrow$  they arrive later. Group delay spread = sum over excited modes.